JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON WEDNESDAY 08th APRIL 2026) **TIME : 3:00 PM TO 6:00 PM**

PHYSICS

TEST PAPER WITH SOLUTION

SECTION-A

26. A new unit (α) of length is chosen such that it is equal to the speed of light in vacuum. What is the distance between Venus and Earth in terms of α units if light takes 6 min. 40 s to cover this distance ?

(1) 200 α (2) 400 α

(3) 300 α (4) 500 α

Sol. $\alpha = 3 \times 10^8$ m

Distance = $(400) \times 3 \times 10^8$ m = 400α

27. Consider the equation $H = \dfrac{x^p \in^q E^r}{t^s}$

Where H = magnetic field; E = electric field; $\in$ = permittivity, x = distance, t = time. The values of p, q, r and s respectively are :

(1) 1, 1, 1, 1 (2) –1, 1, 2, 1

(3) 1, –1, –2, 1 (4) –1, –2, –2, 1

Sol. As per the question, considering H as magnetic field

$$[H] = \frac{MLT^{-2}}{LT^{-1}IT} = MI^{-1}T^{-2}$$

$$[E] = \frac{MLT^{-2}}{IT} = MLI^{-1}T^{-3}$$

$$[\in] = \frac{I^2T^2}{MLT^{-2}L^2} = M^{-1}L^{-3}I^2T^4$$

$MI^{-1}T^{-2} = M^{-q+r} L^{p-3q+r} I^{2q-r} T^{4q-3r-S}$

$-q + r = 1$

$P - 3q + r = 0$

$2q - r = -1$

$4q - 3r - S = -2$

Solving we get

$q = 0, r = 1$

$p = -1, S = -1$

No option matching

Considering H as magnetic field intensity

$H = \dfrac{B}{\mu_0} = Ni \Rightarrow$ current per unit length $\Rightarrow IL^{-1}$

$-q + r = 0$ $q = 1$

$p - 3q + r = -1$ $r = 1$

$2q - r = 1$ $p = +1$

$4q - 3r - s = 0$ $s = +1$

Option (1) is matching

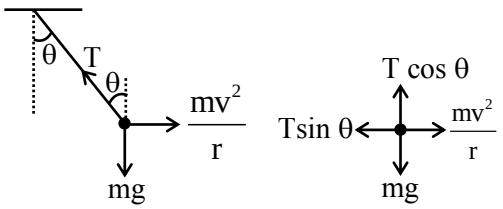
Note : As per the question if we consider H as magnetic field, no option is matching but if we consider H as magnetic field intensity then option (1) is matching.

28. A car moving with a speed of 54 km/h takes a turn of radius 20 m. A simple pendulum is suspended from the ceiling of the car. Determine the angle made by the string of the pendulum with the vertical during the turning. (Take g = 10 m/s^2)

(1) $\tan^{-1}(0.5)$ (2) $\tan^{-1}(0.75)$

(3) $\tan^{-1}(1.125)$ (4) $\tan^{-1}(0.25)$

Sol.



$\tan\theta = \dfrac{v^2}{rg}$ $\left(v = 54 \times \dfrac{5}{18} = 15 \text{m/s}\right)$

$\tan\theta = \dfrac{225}{20 \times 10}$

$\tan\theta = \dfrac{9}{8}$

$\theta = \tan^{-1}(1.125)$

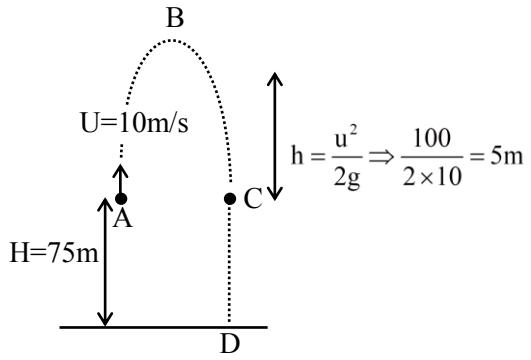
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29. A gas balloon is going up with a constant velocity of 10 m/s. When this balloon reached a height of 75 m, a stone is dropped from it and balloon keeps moving up with the same velocity. The height of the balloon when the stone hits the ground is _____ m. (Take g = 10 m/s^2)

(1) 85 (2) 150

(3) 129 (4) 125

Sol.



$-75 = 10 \times t - \frac{1}{2} \times 10 \times t^2$

$t^2 - 2t - 15 = 0$

$(t-5)(t+3) = 0 \Rightarrow t = 5$

Height = 75 + 10 × 5 = 125 m

30. A thin biconvex lens is prepared from the glass (μ = 1.5) both curved surfaces of which have equal radii of 20 cm each. Left side surface of the lens is silvered from outside to make it reflecting. To have the position of image and object at the same place, the object should be placed, from the lens at a distance of _____ cm.

(1) 10 (2) 12.5

(3) 13 (4) 13.5

Sol. Combination of lens + mirror

$f_m = \frac{R}{2} = -10\text{cm}$; $f_L = \frac{R}{2(\mu-1)} = 20\text{cm}$

$\frac{1}{f} = \frac{1}{f_m} - \frac{2}{f_L}$

$\frac{1}{f} = \frac{-1}{10} - \frac{2}{20} = -\frac{1}{5}$

$f = -5$ cm

To form image at the position of object, it should be placed at center of curvature of mirror + lens combination.

$\therefore u = 2f = 2 \times 5 = 10$ cm

31. Two identical bodies, projected with the same speed at two different angles cover the same horizontal range R. If the time of flight of these bodies are 5 s and 10 s, respectively, then the value of R is _____ m. (Take g = 10 m/s^2)

(1) 250 (2) 25

(3) 500 (4) 125

Sol. For range to be same, angle of projection must be complementary angles.

$T_1 = \frac{2u\sin\theta}{g} = 10$, $T_2 = \frac{2u\cos\theta}{g} = 5$

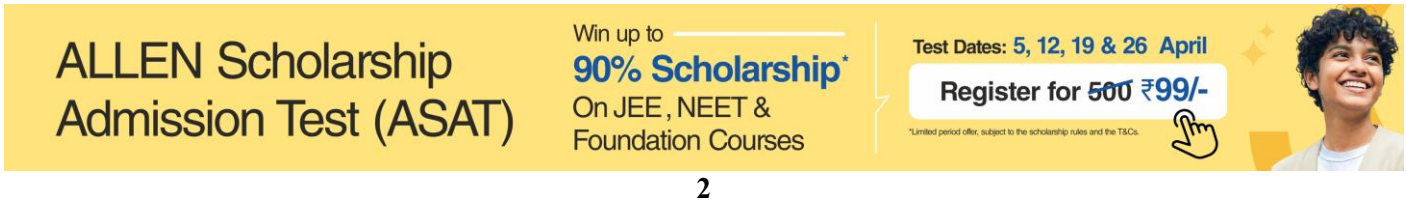
$\text{Range} = \frac{2(u\sin\theta)(u\cos\theta)}{g}$

$= \frac{2(50)(25)}{10}\text{m} = 250\,\text{m}$

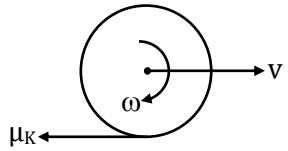
32. A solid cylinder having radius R and length L is slipping on a rough horizontal plane. At time t = 0 the cylinder has a translational velocity v_0 = 49 m/s, perpendicular to its axis and a rotational velocity v_0 /4R about the centre. The time taken by the cylinder to start rolling is _____ seconds. (coefficient of kinetic friction μ_K = 0.25 and g = 9.8 m/s^2)

(1) 15 (2) 5

(3) 10 (4) 7.5



Sol.



$$a = \frac{\mu_K mg}{m} = \mu_K g$$

$$\tau = I\alpha \Rightarrow \mu_K mgR = \frac{MR^2}{2} \times \alpha$$

$$\Rightarrow \alpha = \frac{2\mu_K g}{R}$$

$$\therefore v = v_0 - \mu_k gt \quad \ldots (i)$$

$$\text{Also, } \omega = \omega_0 + \frac{2\mu_K gt}{R} \quad \ldots (ii)$$

$$\frac{v}{R} = \frac{v_0}{4R} + \frac{2\mu_k gt}{R}$$

Solving Eq. (i) & (ii)

$$\frac{v_0}{4} + 2\mu gt = v_0 - \mu_k gt$$

$$3\mu gt = v_0 - \frac{v_0}{4}$$

$$3\mu gt = \frac{3v_0}{4}$$

$$\Rightarrow t = \frac{v_0}{4\mu_K g} = \frac{49}{4 \times \frac{1}{4} \times 9.8} = 5 \text{sec.}$$

33. A liquid of density 600 kg/m^3 flowing steadily in a tube of varying cross-section. The cross-section at a point A is 1.0 cm^2 and that at B is 20 mm^2. Both the points A and B are in same horizontal plane, the speed of the liquid at A is 10 cm/s. The difference is pressures at A and B points is _____ Pa.

(1) 18 (2) 144

(3) 36 (4) 72

Sol. Using continuity equation,

$$A_1V_1 = A_2V_2$$

$$1 \times 10 = \frac{1}{5} \times V_2$$

$$\Rightarrow V_2 = 50 \text{ cm/s}$$

Using Bernoulli's theorem,

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$$

$$P_1 - P_2 = \frac{1}{2}\rho\left[V_2^2 - V_1^2\right]$$

$$= \frac{1}{2} \times 600\left[(50)^2 - (10)^2\right] \times 10^{-4} = 72 \text{ Pa.}$$

34. A spherical liquid drop of radius R acquires the terminal velocity v_1 when falls through a gas of viscosity η. Now the drop is broken into 64 identical droplets and each droplet acquires terminal velocity v_2 falling through the same gas. The ratio of terminal velocities v_1/v_2 is _____.

(1) 4 (2) 0.25

(3) 32 (4) 16

Sol. Using volume conservation,

$$\frac{4}{3}\pi R_1^3 = 64 \times \frac{4}{3}\pi R_2^3 \Rightarrow R_2 = \frac{R_1}{4}$$

Terminal velocity $v \propto R^2$

$$\frac{v_1}{v_2} \Rightarrow \frac{R_1^2}{R_2^2} \Rightarrow (4)^2 = 16$$

35. One mole of diatomic gas having rotational modes only is kept is a cylinder with a piston system. The cross-section area of the cylinder is 4 cm^2. The gas is heated slowly to raise the temperature by 1.2 °C during which the piston moves by 25 mm. The amount of heat supplied to the gas is _____J. (Atmospheric pressure = 100 kPa, R = 8.3 J/mol.K) (Neglect mass of the piston)

(1) 24.8 (2) 25

(3) 15.04 (4) 29.98

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Sol. **Ist Method**

$Q = \Delta U + W$

$Q = \frac{f}{2}nR\Delta T + nR\Delta T = \left(\frac{f}{2}+1\right)nR\Delta T$

if $f = 2 \Rightarrow Q = 2nR\Delta T \Rightarrow 2 \times 1 \times 8.3 \times 1.2 = 19.92$

(No option is matching)

if $f = 5 \Rightarrow Q = 3.5nR\Delta T$

$\Rightarrow 3.5 \times 1 \times 8.3 \times 1.2 = 34.86$

(No option is matching)

Method-2

$Q = \Delta U + W$

$Q = \frac{f}{2}nR\Delta T + P\Delta V$

if $f = 2$

$\Rightarrow Q = nR\Delta T + P\Delta V$

$= 1 \times 8.3 \times 1.2 + 10^5 \times 25 \times 10^{-3} \times 4 \times 10^{-4}$

$= 10.96$

(No option is matching)

if $f = 5$

$\Rightarrow Q = 2.5nR\Delta T + P\Delta V$

$2.5 \times 1 \times 8.3 \times 1.2 + 10^5 \times 25 \times 10^{-3} \times 4 \times 10^{-4}$

$= 25.9$

(closest is 25)

Note : As $P\Delta V \neq nR\Delta T$, given state is inconsistent

Hence we can opt the most suitable option as (2).

36. Initial pressure and volume of a monoatomic ideal gas are P and V. The change in internal energy of this gas in adiabatic expansion to volume $V_{final} = 27\ V$ is _____ J.

(1) $-2PV\left(3\sqrt{3}-1\right)$ (2) $\frac{4}{3}PV$

(3) $-\frac{4}{3}PV$ (4) $\frac{3}{4}PV$

Sol. $P_1V_1^{\gamma} = P_2V_2^{\gamma}$

$\gamma = \frac{5}{3}$

$PV^{5/3} = P_2 \times (27V)^{5/3}$

$P_2 = \frac{P}{(27)^{5/3}} = \frac{P}{3^5}$

$\Delta U \Rightarrow nC_v\Delta T = \frac{nR\Delta T}{(\gamma-1)}$

$= \frac{P_2V_2 - P_1V_1}{(\gamma-1)}$

$\Rightarrow \frac{\frac{P}{3^5} \times 3^3.V - PV}{\frac{5}{3}-1} \Rightarrow \frac{-PV\left\{1-\frac{1}{9}\right\}}{\frac{2}{3}}$

$= -PV \times \frac{8}{9} \times \frac{3}{2}$

$\Rightarrow -\frac{4}{3}PV$

37. The frequency of oscillation of a mass m suspended by spring is v_1. If the length of the spring is cut to half, the same mass oscillates with frequency v_2. The value of $\frac{v_2}{v_1}$ is _____.

(1) 1 (2) 2

(3) $\sqrt{2}$ (4) $\sqrt{3}$

Sol. For a spring, $k\ell$ = constant

$k_1\ell = k_2\frac{\ell}{2} \Rightarrow k_2 = 2k_1$

$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

$\Rightarrow \frac{f_1}{f_2} = \sqrt{\frac{k_1}{k_2}}$

$\Rightarrow \frac{f_1}{f_2} = \sqrt{\frac{1}{2}} \Rightarrow \frac{f_2}{f_1} = \sqrt{2}$

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38. A monochromatic source of light operating at 15 kW emits 2.5×10^{22} photons/s. The region of an electromagnetic spectrum to which the emitted electromagnetic radiation belongs to _____. (Take $h = 6.6 \times 10^{-34}$ J.s and $c = 3 \times 10^{8}$ m/s).

(1) Microwave (2) Infrared

(3) Visible (4) Ultraviolet

Sol. $\frac{hC}{\lambda} = \frac{15\times10^{3}}{2.5\times10^{22}}$

$\therefore \lambda = \frac{6.6\times10^{-34}\times3\times10^{8}\times2.5\times10^{22}}{15\times10^{3}}$

$\lambda = 3.3 \times 10^{-7}$

39. A current carrying circular loop of radius 2 cm with unit normal $\hat{n} = \frac{\hat{k}+\hat{i}}{\sqrt{2}}$ is placed in a magnetic field, $\vec{B} = B_0(3\hat{i}+2\hat{k})$. If $B_0 = 4 \times 10^{-3}$ T and current $I = 100\sqrt{2}A$, the torque experienced by the loop is _____ Wb.A. ($\pi = 3.14$)

(1) $16\times10^{-5}\hat{k}$ (2) $5024\times10^{-7}\hat{k}$

(3) $5024\times10^{-7}\hat{i}$ (4) $5024\times10^{-7}\hat{j}$

Sol. $\vec{\tau} = \vec{M}\times\vec{B}$

$\Rightarrow 100\sqrt{2}\times3.14\times4\times10^{-4}\left[\left(\frac{\hat{k}+\hat{i}}{\sqrt{2}}\right)\times\left(3\hat{i}+2\hat{k}\right)\right]B_0$

$\Rightarrow 16\times3.14\times\left[+3\hat{j}-2\hat{j}\right]\times10^{-5}$

$\Rightarrow 50.24\times10^{-5}\hat{j}$

40. A 30 cm long solenoid has 10 turns per cm and area of 5 cm^2. The current through the solenoid coil varies from 2 A to 4 A is 3.14 s. The e.m.f. induced in the coil is $\alpha \times 10^{-5}$ V. The value α is _____.

(1) 60 (2) 12

(3) 120 (4) 34

Sol. $\varepsilon = -\frac{Ldi}{dt} = -\mu_0 n^2 A\ell\frac{di}{dt}$

$= -4\pi\times10^{-7}\times10^{6}\times5\times10^{-4}\times0.3\times\frac{2}{\pi}$

$\Rightarrow -12\times10^{-5}$

41. Two point charges $q_1 = 3\mu C$ and $q_2 = -4\mu C$ are placed at points $(2\hat{i}+3\hat{j}+3\hat{k})$ and $(\hat{i}+\hat{j}+\hat{k})$ respectively. Force on charge q_2 is _____ N.

$\left(\text{Take } \frac{1}{4\pi\epsilon_0} = 9\times10^{9}\text{ SI Units}\right)$

(1) $(12\hat{i}+24\hat{j}+24\hat{k})\times10^{-3}$

(2) $(4\hat{i}+8\hat{j}+8\hat{k})\times10^{-3}$

(3) $(3\hat{i}+6\hat{j}+6\hat{k})\times10^{-3}$

(4) $(-4\hat{i}-8\hat{j}-8\hat{k})\times10^{-3}$

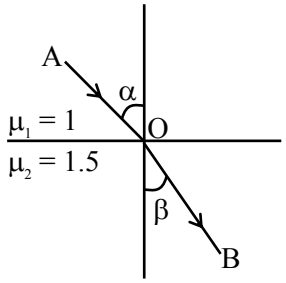
Sol. $\vec{r} = \vec{r}_2 - \vec{r}_1 \Rightarrow -\hat{i}-2\hat{j}-2\hat{k}$

Force $\Rightarrow \frac{kq_1q_2}{r^3}\vec{r}$

$\Rightarrow \frac{9\times10^{9}\times(3)(+4)\times10^{-12}}{27}\left(\hat{i}+2\hat{j}+2\hat{k}\right)$

$\Rightarrow 4\times10^{-3}\left[\hat{i}+2\hat{j}+2\hat{k}\right]$

42. Light ray incident along a vector $\overrightarrow{AO}\left(\overrightarrow{AO} = 2\hat{i}-3\hat{j}\right)$ emerges out along vector $\overrightarrow{OB}\left(\overrightarrow{OB} = C\hat{i}-4\hat{j}\right)$ as shown in the figure below. The value of C is _____.



(1) 1.6 (2) 0.16

(3) 11.6 (4) 16

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Sol. $1\left(AO\times\hat{j}\right)\Rightarrow\left(OB\times\hat{j}\right)\times1.5$

$$\frac{2\hat{k}}{\sqrt{13}}=\frac{3}{2}\left[\frac{C\hat{k}}{\sqrt{C^2+16}}\right]$$

$$\therefore\frac{4}{13}=\frac{9C^2}{4\left(C^2+16\right)}\Rightarrow16C^2+256=117C^2$$

$256 = 101\ C^2$

$C = 1.59 \approx 1.6$

43. K_1 and K_2 be the maximum kinetic energies of photoelectrons emitted from a surface of a given material for the light of wavelength λ_1 and λ_2, respectively. If $\lambda_1 = 2\lambda_2$ then the work function of material is given by :

(1) $K_2 + 2K_1$ (2) $2K_2 - K_1$

(3) $K_1 - 2K_2$ (4) $K_2 - 2K_1$

Sol. $2K_1=\frac{2hC}{\lambda_1}-2\phi$ … (i)

$K_2=\frac{2hC}{\lambda_1}-\phi$ … (ii)

Solving (i) & (ii), we get

$K_2 - 2K_1 = \phi$

44. Two radioactive substances A and B of mass numbers 200 and 212 respectively, shows spontaneous α-decay with same Q value of 1 MeV. The ratio of energies of α-rays produced by A and B is _____.

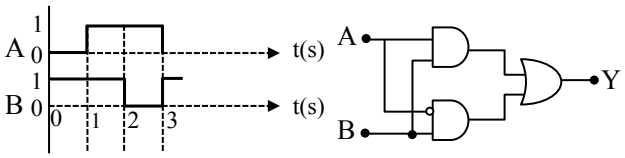
(1) $\frac{2548}{2650}$ (2) $\frac{2706}{2646}$

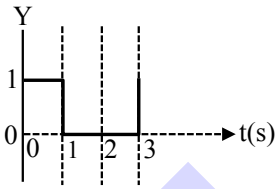
(3) $\frac{2597}{2600}$ (4) $\frac{2862}{2499}$

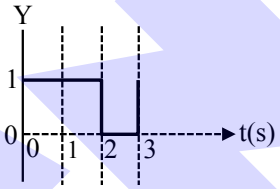
Sol. $k_\alpha=\left(\frac{A-4}{A}\right)Q$

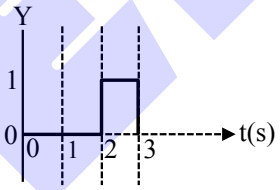
$$\therefore\frac{k_{\alpha_1}}{k_{\alpha_2}}=\frac{196}{200}\times\frac{212}{208}=\frac{2597}{2600}$$

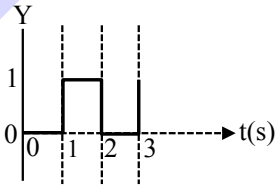
45. The output Y for the given inputs A and B to the circuit is :

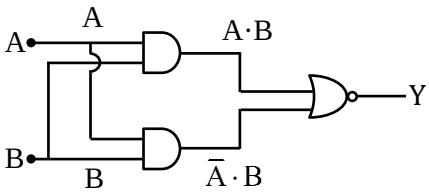


(1)


(2)


(3)


(4)


Sol.


$A\cdot B+\bar{A}\cdot B\Rightarrow\left(A+\bar{A}\right)\cdot B=B$

Note : In lower gate, input is taken from A as $\bar{A}$ due to node given in the figure.

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SECTION-B

46. A parallel plate capacitor is having separation between plates 0.885 mm. It has a capacitance of 1 μF when the space between the plates is filled with an insulating material of resistivity 1×10^{13} Ωm and resistance 17.7×10^{14} Ω. Relative permittivity of the insulating material is $\alpha \times 10^{7}$. The value of α is _____.

(Take permittivity of free space = 8.85×10^{-12} F/m)

Sol. $RC = \rho K \in_0$

$$K = \frac{RC}{\rho \varepsilon_0} = \frac{17.7 \times 10^{14} \times 10^{-6}}{10^{13} \times 8.85 \times 10^{-12}} = 2 \times 10^{7}$$

47. Some distance star is to be observed by some telescope of diameter of objective lens a, at an angular resolution of 3.0×10^{-7} radian. If the wavelength of light from the star reaching the telescope is 500 nm, the minimum diameter of the objective lens of the telescope is _____ cm. (Nearest integer)

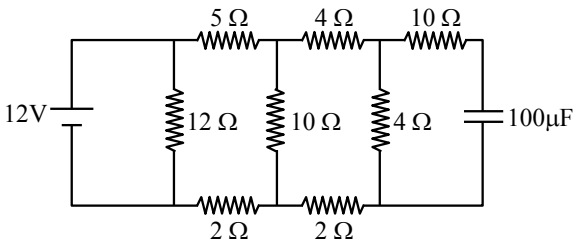
Sol. $\theta = \frac{1.22\lambda}{b}$

$$3 \times 10^{-7} = \frac{1.22 \times 5 \times 10^{-7}}{b} \Rightarrow b = 203.33 \text{cm}$$

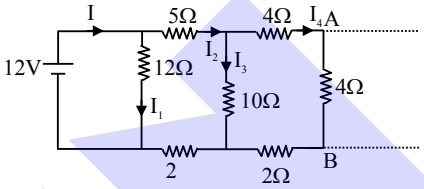
48. A 5 mg particle carrying a charge of $5\pi \times 10^{-6}$ C is moving with velocity of $(3\hat{i} + 2\hat{k}) \times 10^{-2}$ m/s in a region having magnetic field $\vec{B} = 0.1\,\hat{k}$ Wb/m^2. It moves a distance of α meter along $\hat{k}$ when it completes 5 revolutions. The value of α is _____.

Sol. $5\left(\frac{2\pi m}{qB}\right)v \Rightarrow \frac{5 \times 2 \times \pi \times 5 \times 10^{-6} \times 2 \times 10^{-2}}{5\pi \times 10^{-6} \times 0.1} = 2\text{m}$

49. The stored charge in the capacitor in steady state of the following circuit is _____ μC.



Sol. At steady state,



$$I = \frac{E}{R_{eq}} = \frac{12}{6} = 2A$$

$$I_1 = I_2 = 1A$$

$$I_3 = I_4 = \frac{1}{2}A$$

$$V_{AB} = \frac{1}{2} \times 4 = 2V$$

Charge on capacitor

$Q = CV_{AB} = 100 \times 2 = 200$ μC

50. Two masses of 3.4 kg and 2.5 kg are accelerated from an initial speed of 5 m/s and 12 m/s, respectively. The distances traversed by the masses in the 5th second are 104 m and 129 m, respectively. The ratio of their momenta after 10 s is $\frac{x}{8}$. The value of x is _____.

Sol. $S = u + \frac{a}{2}(2n-1) \Rightarrow 104 = 5 + \frac{a}{2}(9) \Rightarrow a = 22 \text{ m/s}^2$

$$129 = 12 + \frac{a}{2} \times 9 \Rightarrow a = 26 \text{ m/s}^2$$

$$\frac{p_1}{p_2} = \frac{(5 + 22 \times 10)3.4}{(12 + 26 \times 10)(2.5)} = \frac{225 \times (3.4)}{272 \times 2.5} = \frac{9}{8}$$

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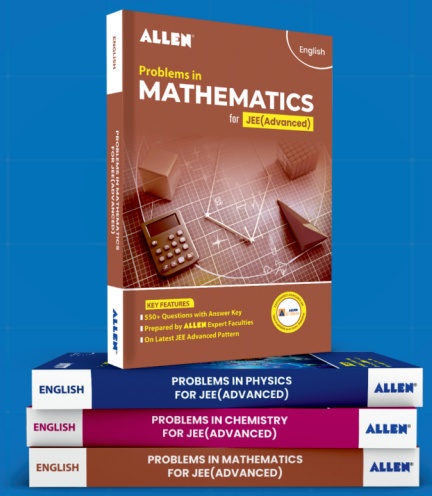


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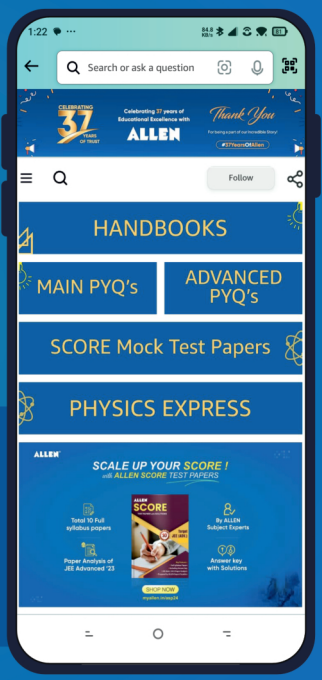
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